

GENIE

Google Gemini Powered Learning Assistant

Project Testing Phase

Verifying functionality, reliability, and response handling

Project Phase Documentation

This document covers the activities, outputs, and considerations for this phase of the Genie project.

1. Testing Objectives

Genie – Project Testing Phase

- Verify that Genie starts and runs without major errors.
- Check that user input is accepted correctly.
- Verify that the application can communicate with the configured Gemini service.
- Confirm that generated responses are displayed correctly.
- Identify and fix errors before the final demonstration.

Aspect	Details
Purpose	Verifying functionality, reliability, and response handling
Project	Genie: Google Gemini Powered Learning Assistant
Technology	Python + Google Gemini integration
Target Users	Students / learners

2. Test Cases

Genie – Project Testing Phase

- Test with a simple factual learning question.
- Test with a request for a step-by-step explanation.
- Test with a request to summarize a topic.
- Test with an empty or invalid input to verify validation.
- Test behavior when the AI service or internet connection is unavailable.
- Test repeated prompts to ensure the interface continues to work correctly.

Key Note: The exact implementation can be adjusted based on the final project code and available Gemini API configuration.

3. Results, Error Handling & Validation

Genie – Project Testing Phase

- Record whether each test case produces the expected application behavior.
- Check that error messages are understandable to the user.
- Verify that API credentials are not displayed in the interface or committed to a public repository.
- Review generated responses for relevance and clarity as part of application evaluation.
- Fix identified issues and repeat important tests after changes.
- Testing output: a validated project version suitable for documentation and demonstration.

Phase Output: A clear, reviewable result that can be carried forward to the next project phase.